Solutions Tutorial 10

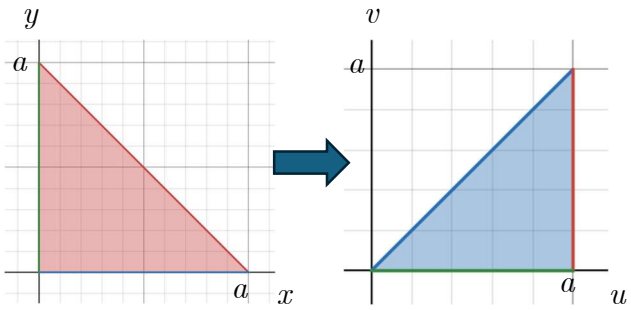
1. By using the change of variable $u = x + y,\ v = x$, evaluate $\iint_{T_a} (x+y)dxdy$, where T_a is the region in the xy-plane bounded by the x and y axes and the line $x + y = a > 0$.

Solution. Given $x = v$ and $u = x + y \Rightarrow y = u - v$. Now

$$J = \det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1 \Rightarrow |J| = 1.$$

$$x = 0 \Rightarrow v = 0;\ \ y = 0 \Rightarrow u = v;\ \ x + y = a \Rightarrow u = a$$

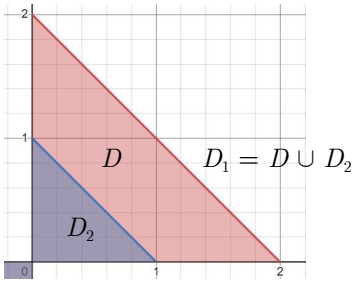
(green to green) (blue to blue) (red to red)



Hence, $\iint_{T_a} (x+y)dxdy = \iint_{T_a'} u \times 1 du dv = \int_0^a \int_0^u u dv du = \int_0^a u^2 du = \frac{1}{3}a^3$.

2. Use the result of Q1 to deduce the value of $\iint_D (x+y)dxdy$, where D is the region in the xy plane bounded by the x and y axes and the lines $x + y = 1$ and $x + y = 2$.

Solution. Let $D_1 = \{(x, y): x + y \le 2,\ x, y \ge 0\}$; $D_2 = \{(x, y): x + y \le 1,\ x, y \ge 0\}$.



$$\iint_D (x+y)dxdy = \iint_{D_1} (x+y)dxdy - \iint_{D_2} (x+y)dxdy = \frac{1}{3}(2^3 - 1^3) = \frac{7}{3}.$$

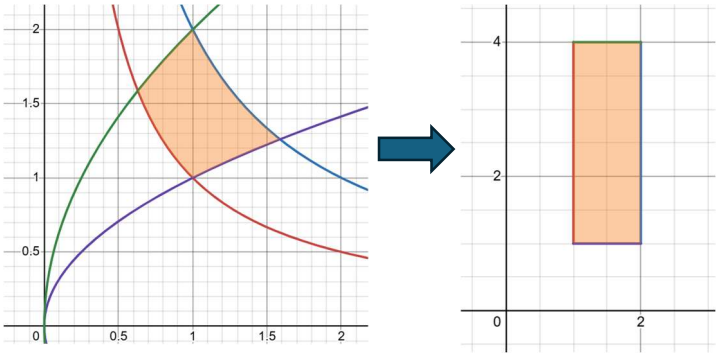
3. By using a suitable substitution, find the area of the region in the xy plane bounded by the curves $xy=1,\ xy=2,\ y^2=x$ and $y^2=4x$.

Solution. Let $u=xy$ and $v=\frac{y^2}{x}$. So $u=\left(\frac{y^2}{v}\right)y=\frac{y^3}{v}\Rightarrow y^3=uv\Rightarrow y=(uv)^{\frac{1}{3}},\ x=\left(\frac{u^2}{v}\right)^{\frac{1}{3}}$.

Then $$J=\det\begin{pmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{pmatrix}=\begin{vmatrix}\frac{2}{3}u^{-1/3}v^{-1/3} & -\frac{1}{3}u^{2/3}v^{-4/3}\\ \frac{1}{3}u^{-2/3}v^{1/3} & \frac{1}{3}u^{1/3}v^{-2/3}\end{vmatrix}=\frac{1}{3v}\Rightarrow |J|=\frac{1}{3|v|}.$$

$u=xy=1,\ v=\frac{y^2}{x}=1$,
(red) (purple)

$u=xy=2, v=\frac{y^2}{x}=4$.
(blue) (green)



$$\text{Area}=\iint_D 1\,dxdy=\iint_{D'}1\times\frac{1}{3v}\,dudv=\frac{1}{3}\int_1^2 du\int_1^4 v^{-1}dv=\frac{1}{3}(1\times\ln 4)=\frac{1}{3}\ln 4.$$

Without using the substitution:

$$\begin{aligned}\int_{\frac{1}{\sqrt[3]{4}}}^{1}\int_{\frac{1}{x}}^{2\sqrt{x}}dydx+\int_{1}^{\sqrt[3]{4}}\int_{\sqrt{x}}^{\frac{2}{x}}dydx&=\int_{\frac{1}{\sqrt[3]{4}}}^{1}(2\sqrt{x}-\tfrac{1}{x})dx+\int_{1}^{\sqrt[3]{4}}(\tfrac{2}{x}-\sqrt{x})dx\\&=\left[\tfrac{4}{3}x^{\frac{3}{2}}-\ln x\right]_{\frac{1}{\sqrt[3]{4}}}^{1}+\left[2\ln x-\tfrac{2}{3}x^{\frac{3}{2}}\right]_{1}^{\sqrt[3]{4}}\\&=\tfrac{4}{3}-(\tfrac{4}{3}(\tfrac{1}{4})^{\frac{1}{2}}+\tfrac{1}{3}\ln 4)+\tfrac{2}{3}\ln 4-\tfrac{2}{3}(4)^{\frac{1}{2}}+\tfrac{2}{3}\\&=\tfrac{1}{3}\ln 4\end{aligned}$$

4. Let $r=\sqrt{x^2+y^2}$. Find the interval of α such that the integral converges.

(a) $\iint_{B_1} r^{-\alpha}dxdy,\ B_1=\{r: r\geq 1\}$ (b) $\iint_{B_2} r^{-\alpha}dxdy,\ B_2=\{r: 0<r\leq 1\}$

(c) $\iint_{B_3}(1-r)^{-\alpha}dxdy,\ B_3=\{r: 0\leq r<1\}$

Solution. (a) Let $D_R=\{(x,y): 1\leq r<R\}, R>1$ and take $R\to\infty$.

When $\alpha\neq 2$: $\iint_{B_1} r^{-\alpha}dxdy=\lim_{R\to\infty}\int_{\theta=0}^{\theta=2\pi}\int_{r=1}^{r=R} r^{-\alpha}rdrd\theta=\lim_{R\to\infty}2\pi\left[\frac{r^{2-\alpha}}{2-\alpha}\right]_1^R=\begin{cases}\infty, & \alpha<2\\ \frac{2\pi}{\alpha-2}, & \alpha>2\end{cases}$.

When $\alpha=2$: $\iint_{B_1} r^{-2}dxdy=\lim_{R\to\infty}\int_{\theta=0}^{\theta=2\pi}\int_{r=1}^{r=R} r^{-1}drd\theta=\lim_{R\to\infty}2\pi\ln R=\infty$.

So, the integral converges when $\alpha>2$ and diverges otherwise.

(b) Let $D_R = \{(x,y) : R < r \le 1\}, R > 0$ and take $R \to 0^+$.

When $\alpha \neq 2$: $\iint\limits_{B_2} r^{-\alpha} dxdy = \lim\limits_{R\to 0} \int\limits_{\theta=0}^{\theta=2\pi} \int\limits_{r=R}^{r=1} r^{-\alpha} r dr d\theta = \lim\limits_{R\to 0} 2\pi \left[\frac{r^{2-\alpha}}{2-\alpha}\right]_R^1 = \begin{cases} \infty, & \alpha > 2 \\ \frac{2\pi}{2-\alpha}, & \alpha < 2 \end{cases}$.

When $\alpha = 2$ (similar to (a)): $\iint\limits_{B_2} r^{-2} dxdy = \lim\limits_{R\to 0} \int\limits_{\theta=0}^{\theta=2\pi} \int\limits_{r=R}^{r=1} r^{-1} dr d\theta = -\lim\limits_{R\to 0} 2\pi \ln R = \infty$.

Therefore, it converges when $\alpha < 2$ and diverges otherwise.

(c) $D_R = \{(x,y) : 0 \le r < R < 1\}$ and take $R \to 1^-$.

$$\iint\limits_{B_3} (1-r)^{-\alpha} dxdy = 2\pi \lim_{R\to 1} \int\limits_{r=0}^{r=R} (1-r)^{-\alpha} r dr .$$

When $\alpha \neq 1, 2$: $\int\limits_{r=0}^{r=R} (1-r)^{-\alpha} r dr = \frac{1}{1-\alpha}\left[-(1-r)^{1-\alpha} r\Big|_0^R + \int\limits_0^R (1-r)^{1-\alpha} dr\right]$

$= \frac{1}{\alpha-1}\left[(1-R)^{1-\alpha} R + \frac{(1-r)^{2-\alpha}}{2-\alpha}\Big|_0^R\right]$

$= \frac{1}{\alpha-1}\left[(1-R)^{1-\alpha} R + \frac{(1-R)^{2-\alpha}}{2-\alpha} - \frac{1}{2-\alpha}\right]$.

Therefore, $\iint\limits_{B_3} (1-r)^{-\alpha} dxdy = 2\pi \lim\limits_{R\to 1} \int\limits_{r=0}^{r=R} (1-r)^{-\alpha} r dr = \begin{cases} \infty, & \alpha > 1, \alpha \neq 2 \\ \frac{2\pi}{(1-\alpha)(2-\alpha)}, & \alpha < 1 \end{cases}$.

When $\alpha = 1$, $\iint\limits_{B_3} (1-r)^{-\alpha} dxdy = 2\pi \lim\limits_{R\to 1} \int\limits_{r=0}^{r=R} \frac{r}{1-r} dr = 2\pi \lim\limits_{R\to 1} \int\limits_{r=0}^{r=R} \left(\frac{1}{1-r} - 1\right) dr$

$= -2\pi \lim\limits_{R\to 1} \left[r + \ln(1-r)\right]_0^R = \infty.$

When $\alpha = 2$, $\iint\limits_{B_3} (1-r)^{-\alpha} dxdy = 2\pi \lim\limits_{R\to 1} \int\limits_{r=0}^{r=R} \frac{r}{(1-r)^2} dr = \infty$ because

$\int\limits_{r=0}^{r=R} \frac{r}{(1-r)^2} dr = \int\limits_{r=0}^{r=R} \frac{r}{(r-2)^2} dr = \left[\ln|r-1| - \frac{1}{r-1}\right]_0^R = \ln|R-1| - \frac{1}{R-1} - 1 \to \infty$ when $R \to 1^-$.

Note that $\int \frac{x}{(x-1)^2} dx = \ln|x-1| - \frac{1}{x-1}$, $\lim\limits_{x\to 0^-} \ln|x| = -\infty$ and $\lim\limits_{x\to 0^-} \frac{1}{x} = -\infty$ but

$\lim\limits_{x\to 0^-} \left(\ln|x| - \frac{1}{x}\right) = \infty$, (because $\lim\limits_{x\to 0^-} \left(\ln|x| - \frac{1}{x}\right) = \lim\limits_{x\to 0^-} \left(\frac{x\ln|x|-1}{x}\right) = \infty$.)

So, $\iint\limits_{B_3} (1-r)^{-\alpha} dxdy = \infty$. Hence, it converges when $\alpha < 1$ and diverges otherwise.

5. Let $r=\sqrt{x^2+y^2}$. Use the integral comparison test to investigate the integrability of the following integrals:

(a) $\iint\limits_{B_1}\ln(x^2+y^2)\,dxdy$, $B_1=\{r:r\geq 1\}$ (b) $\iint\limits_{B}\frac{1}{x^4+y^4}\,dxdy$, $B=\{r:0\leq r\leq 1\}$

(c) $\iint\limits_{D}\frac{x-y}{x^2+y^2}\,dxdy$, $D=\{(x,y):\left|x\right|<1,\left|y\right|<1\}$

Solution. (a) Consider $\iint_{B'}\ln\left(x^2+y^2\right)dxdy$ where $B'=\{r:r\geq e\}$. Since $r\geq e$, it follows that $\ln\left(x^2+y^2\right)=\ln r^2=2\ln r\geq 2\ln e=2=2r^{-0}$. Note that we could show that $2\iint_{B'}r^0dxdy$ diverges, therefore $\iint_{B'}\ln\left(x^2+y^2\right)dxdy$ diverges. Now since $B'\subset B_1$, we conclude that $\iint_{B_1}\ln\left(x^2+y^2\right)dxdy$ is not integrable.

(b) By applying the triangle inequality:

$$\left|x^4+y^4\right|\leq\left|x\right|^4+\left|y\right|^4=x^4+y^4\leq 2r^4\Rightarrow\frac{1}{x^4+y^4}\geq\frac{1}{2r^4},\ (x,y)\neq(0,0).$$

Then use the fact in Q4(b) and $\iint\limits_{B}\frac{1}{x^4+y^4}\,dxdy\geq\frac{1}{2}\iint\limits_{B_2}\frac{1}{r^4}\,dxdy$, we conclude that $\iint\limits_{B}\frac{1}{x^4+y^4}\,dxdy$ is not integrable as $\iint\limits_{B_2}r^{-\alpha}dxdy$ diverges when $\alpha\geq 2$.

(c) Note that $D=\{(x,y):\left|x\right|<1,\left|y\right|<1\}\subset B''=\{r:0\leq r\leq\sqrt{2}\}$. By applying the triangle inequality: $\left|x-y\right|\leq\left|x\right|+\left|y\right|\leq 2r\Rightarrow\frac{\left|x-y\right|}{r^2}\leq\frac{2}{r}$, $(x,y)\neq(0,0)$. Then use the fact that $D\subset B''$ and $\iint_{D}\frac{\left|x-y\right|}{x^2+y^2}dxdy\leq 2\iint_{B''}\frac{1}{r}dxdy$ to conclude that $\iint_{D}\frac{x-y}{x^2+y^2}dxdy$ is integrable.

6. Evaluate the integrals

(a) $\iint\limits_D e^{-x-y}dxdy\,,\ D=\{(x,y):0\le x\le 1, y\ge 0\}$

(b) $\iint\limits_D x^{-2}y^{-2}dxdy\,,\ D=\{(x,y):x\ge 1, 1\le y\le x^2\}$

(c) $\iint\limits_D e^{-x-y}dxdy\,,\ D=\{(x,y):x,y\ge 0\}$

Solution. (a) $\iint\limits_D e^{-x-y}dxdy=\lim\limits_{Y\to\infty}\int\limits_0^Y\int\limits_0^1 e^{-x-y}dxdy$

$$=\lim_{Y\to\infty}\int_0^Y e^{-y}dy\int_0^1 e^{-x}dx=(1-e^{-1})\lim_{Y\to\infty}(1-e^{-Y})=1-e^{-1}$$

(b) $\iint\limits_D (xy)^{-2}dxdy=\lim\limits_{X\to\infty}\int\limits_1^X\int\limits_1^{x^2}(xy)^{-2}dydx=-\lim\limits_{X\to\infty}\int\limits_1^X\left[x^{-2}y^{-1}\right]_1^{x^2}dx$

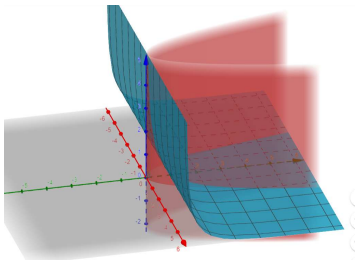
$$=\lim_{X\to\infty}\int_1^X(x^{-2}-x^{-4})dx=\lim_{X\to\infty}\left[-x^{-1}+\tfrac{1}{3}x^{-3}\right]_1^X$$

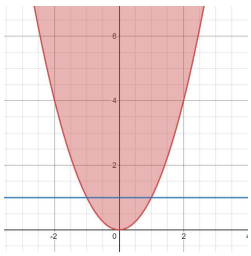
$$=\lim_{X\to\infty}\left((-X^{-1}+\tfrac{1}{3}X^{-3})-(-1+\tfrac{1}{3})\right)=\tfrac{2}{3}$$

(c) $\iint\limits_D e^{-x-y}dxdy=\lim\limits_{R\to\infty}\int\limits_0^R e^{-x}dx\lim\limits_{R\to\infty}\int\limits_0^R e^{-y}dy=\left[\lim\limits_{R\to\infty}(1-e^{-R})\right]^2=1$

7. Evaluate the integral $\iint\limits_D y^{-\frac{1}{2}}e^{-y}dxdy\,,\ D=\{(x,y):y\ge x^2\}\,.$

Solution. We have an unbounded region (above $y=x^2$), and the singularity of the function is at $y=0$. Consider only the first octant (and multiply the answer by 2).





Let $D_R=D_1\cup D_2\,, D_1=\{(x,y):0\le x\le\sqrt{y}, R_1\le y\le 1, R_1>0\}$, and

$D_2=\{(x,y):0\le x\le\sqrt{y}, 1\le y\le R_2, R_2>1\}\,.$

$$\iint\limits_{D_1} y^{-\frac{1}{2}}e^{-y}dxdy=\lim_{R_1\to 0}\int_{R_1}^1\int_0^{\sqrt{y}}\frac{dxdy}{e^y\sqrt{y}}=\lim_{R_1\to 0}\int_{R_1}^1\left[\frac{x}{e^y\sqrt{y}}\right]_0^{\sqrt{y}}dy=\lim_{R_1\to 0}\int_{R_1}^1 e^{-y}dy=\lim_{R\to 0}(e^{-R_1}-e^{-1})=1-e^{-1}$$

$$\iint\limits_{D_2} y^{-\frac{1}{2}}e^{-y}dxdy=\lim_{R_2\to\infty}\int_1^{R_2}\int_0^{\sqrt{y}}\frac{dxdy}{e^y\sqrt{y}}=\lim_{R_2\to\infty}\int_1^{R_2}\left[\frac{x}{e^y\sqrt{y}}\right]_0^{\sqrt{y}}dy=\lim_{R_2\to\infty}\int_1^{R_2}e^{-y}dy=\lim_{R\to\infty}(e^{-1}-e^{-R_2})=e^{-1}$$

Hence, $\iint\limits_D y^{-\frac{1}{2}}e^{-y}dxdy=2\left(1-e^{-1}+e^{-1}\right)=2\,.$